

Image Recognition to Aid Fire Evacuations

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Abstract

This project is about indoor demographic AI for fire prevention, which uses computer deep learning to implement computer vision and applies this functionality to the project through some programming. There are now many proven models like this project in the world, such as people counting and traffic flow counting through computer vision. But the major difference between this project and these projects is that the number of people in a building is counted. It needs to track the movement of different people in the building in real time to determine the number of people in different areas in real time. This is an interesting and socially beneficial project because it is not just about writing code. The project is very interactive with the developers, such as the construction and selection of the environment and the production of the dataset. The computer vision algorithm needs to use the dataset to train the weight file, which corresponds to training the accuracy of recognition and which objects can be recognised. The training results also need to be evaluated for testing, this includes using different videos, images to test the results. Once the object detection is complete the tracking algorithm is used to achieve real-time tracking of the object. The code section focuses on the statistical counting of numbers and the determination of trigger statistics, and the project has developed some features to help improve ease of use and a simple and clear interactive interface.

[Keyword: object recognition, object tracking, headcount, indoor environment]

Preface

The significance of this project is to aid in fire evacuation using computer vision technology, a study with practical applications. With urbanisation and population growth, fire evacuation is increasingly becoming a critical issue, especially in large public spaces and high-rise buildings. This project uses image recognition technology that can automatically detect the number of people in different areas and provide valuable information as an aid to fire evacuation. This will help to reduce casualties and property damage and improve evacuation efficiency, making it very interesting research.

Because computer vision is a cross-disciplinary and comprehensive discipline, the project involves knowledge from a number of different fields. The project also used a number of different open-source codes as a root for development and some design inspiration from comparing and analysing similar products. Many challenges and obstacles were encountered in the development of the project, most of which have been overcome, but there are still some issues that have not been fully resolved, such as the CCTV call permissions and the optimal location of the bound box trigger condition in the overlapping area.

It was found that it was feasible to use computer vision to count the number of people in the room, and the project could not be integrated into the surveillance system for the time being due to CCTV access issues, but after testing with a real video shared by a family member at their school, it was concluded that this direction was feasible. The computerised statistics are many times faster than manual statistics. Of course, there have been cases where the statistics were inaccurate, mostly due to the slight overlapping area of the trigger points and the overlapping of the portraits.

I believe that in the future these problems will be solved. If this project is developed in the future, it will contribute to the development of intelligent security and the application of image recognition technology. By using image recognition technology to provide more accurate information for fire evacuation, the project has improved the intelligence of disaster risk management, which is an important contribution to the development of intelligent safety. And the project also explores how image recognition technology can be applied to practical scenarios, so it also provides useful experience and reference for future applications.

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1. Introduction

1.1. Background introduction

This project is mainly about the field of artificial intelligence, specifically, object detection in computer vision. Object detection is an important field of artificial intelligence technology, which is widely used in computer vision, robotics, and intelligent transportation. In the field of computer vision, object detection can be used to identify objects in images or videos, and determine their types, positions, motion tracks and other information. This technology can be used in image classification, target tracking, video summarization and other applications. In the field of robotics, object detection can be used to help robots understand the surrounding environment and carry out path planning and obstacle avoidance based on environmental information. In the field of intelligent transportation, object detection can be used to identify vehicles, pedestrians, and obstacles on the road, thus providing support for safe driving. In general, object detection is an important field in artificial intelligence technology.

Currently, most object detection is performed on static objects, or on statistical data of pedestrian and vehicle flows, or use in the industrial direction. Although there are similar models, they are not applied to the direction of indoor fire prevention. This project is applicable to such situations. Therefore, when people in the building are evacuated, the number of people in each room can be known. The system will know whether everyone is evacuated and can provide a count for each monitored room. This model will also have great expansion space in the future. It can judge the fire point and fire spread direction according to its own functions, and even add some other algorithms to judge other things. For example, whether the site is safe and whether there are any dangerous actions. This project will use some open-source software to help complete this project. In this project, there are many ways to achieve project goals, such as the mainstream Mask-RCNN and YOLO. The project will also apply many basic computer technologies, such as the coding of the number of people, the call of the model and the code encapsulation. It also includes the underlying logic of digital image and the construction of computer environment, image format and image detection methods (the essence of video is the rapid display of multiple images, so the same is true of video). There are also some complex computer technologies, such as computer vision and artificial intelligence. Computers originally have no vision, but the computer can recognize features through feature detection and image feature filtering so that the computer can see objects.

1.2. Tenet and goals

The objective of this project is to develop a dependable monitoring system for indoor fire evacuation that is capable of precisely recognizing and counting indoor persons and giving correct information on personnel evacuation in an emergency. In order to meet specific objectives and criteria, the goal of this project was to design and create an effective object recognition and tracking system and apply it to a realistic interior monitoring situation. Also, in

order to build this project, it is necessary towards becoming knowledgeable in the use of pertinent open-source software and technologies in order to successfully put them to use in achieving the project's objectives. The project must ultimately be finished in its totality, including the system design, the writing of the code, testing, implementation, and documentation. This project also serves as a personal graduation project for me, allowing me to showcase my expertise in artificial intelligence and gain a thorough grasp of how object detection is used in practice.

1.3. Method description

Anaconda was used to create a flexible environment that made it simple to test various models and sets of data. Valid photos were gathered for dataset pre-processing, dataset annotation, and examination of confidence in object detection. The YOLOv5 open-source code was retrieved from GitHub. The models were then trained, briefly tested, and assessed. The best weighting model was selected to achieve high accuracy target detection. After that, DeepSort and I worked together to create target tracking and amount detection features. In order to make it easier for users to interact with the interface, the code was finally put into a GUI form. There are four distinct phases of development for the project, each of which is crucial. Each of the four phases contains different tasks that are one of the subjects in the task, which means they are independent but can be bridged together.

1.4. Development Methods and Framework

1.4.1. Software development methods

For software development, this project used the incremental model development approach. The incremental approach breaks down the entire software development process into many small incremental parts, each of which includes the ability to test, evaluate, and receive feedback. After completing the first incremental part the next incremental part is started and the cycle continues until the entire software development process is completed.

The key advantages of the incremental model are its speedy software release, earlier client feedback gathering, and programme enhancement. The incremental approach aids in lowering the process' total cost and risk of developing software. The incremental model stresses incremental additions or changes to software functionality to finish, even while each incremental component is very tiny and hence easy to test and debug. The detector, tracker, statistical algorithm, and code wrapper are the initial four models that make up this project. The identifier is followed by the tracker and statistical algorithm. Although they must all be altered and coded independently, they are articulated with one another. As a result, this project is a great fit for this development paradigm.

However, the incremental model also has significant drawbacks, including the fact that integrating incremental pieces can be difficult and requires higher management and coordination abilities, and that changes made inside incremental sections may affect other

sections. Being upfront about the negative aspects of this project's development strategy is crucial. Example: As a result of programme bridging, versioning issues arise. The memory of the earlier components developing will become blurred if a project is developed for a protracted period. It is important to pay closer attention to how the previously built parts will interact with any software or dependency sources that will be used in the next portions as they are being developed.

1.4.2. Requirements analysis

In the requirements analysis piece, this project will assess requirements into two broad categories. Functional requirements and non-functional requirements. This and in this case, Moscow will be used to analyse the need for functionality.

A software requirements classification technique called Moscow will help the project team to determine which features and functions are necessary and which are desirable but not essential. The UK's National Computing Centre was the first organization to come up with Moscow. It is made up of four-letter abbreviations that stand for the following:

M (must have): Essential features and functions.

S (should have): Features and functions that should be included.

C (could have): More features and functions.

W (wouldn't have): Missing features and functions.

The benefit of the Moscow technique is that it helps teams to quickly identify which features are most critical, allowing them to start developing them early in the project. As a result, clients get their money's worth earlier and project time and resources can be better planned.

2. Literature review

2.1. Literature introduction

This article will select some books, Google Scholar and some Influential journals to collect and read useful literature. At the same time, this project involves open source code, which comes from GitHub. The detailed database and environment will be listed at the end of the article ([Appendix 11.1.6](#)). The potential of computer vision for number crunching will be covered in depth in this article, along with an explanation of the field's fundamentals and its use. The design and development process, as well as the analysis of the end outcomes, will also be covered in this article. One of the areas of artificial intelligence development is computer vision. A multidisciplinary field called computer vision studies how to increase the flexibility and automation of manufacturing by teaching computers to comprehend two-dimensional digital images or video more deeply. In dangerous job settings where manual labour is not appropriate or when human eyesight struggles to satisfy the requirements, computer vision will also be used. The goal of this field of study, from an engineering perspective, will be to create a completely automated bionic vision system that is based on the human visual system. Some people believe that machine vision can be of great help to human society and development, but others believe that the market is hyping the capabilities of machine vision. The truth, however, is that machine vision has now moved far beyond its early stages. When Google started investing in this technology, machine vision was like a two- or three-year-old child, whereas now it is a young child learning in kindergarten. Although the two stages don't look very different, a closer comparison shows how much it has progressed, just as it has learned to communicate, learned to run, and at the same time made subtle changes in its ideology. This is the stage of change.

2.2. Main body

2.2.1. Historical development

By replicating the human visual recognition system, individuals have attempted to utilise computers in the past to detect images. This is a very tough undertaking since human vision is one of the hardest sensory cognitive systems for the brain to duplicate. Humans see 3D items in the real world while seeing 2D representations of the same objects in photographs.

Early years:

People was initially written to compare a target image with the database to determine whether it contained roses or a particular target image after people first added roses or, for example, several target images to the database. However, the main drawback of this programme was that it could not distinguish every single image that wasn't in the database. As a result, there is a

great demand for a network that can recognise and identify spatial features in photos automatically. (Bhatt, D. et al.2021)

1960s-70s:

In his research paper BLOCK WORLD from 1963, computer scientist Larry Roberts—known as the inventor of computer vision—illustrated the potential for deriving 3D geometric information from 2D perspective views of objects. This was regarded as the first significant development in computer vision. After this work, a large number of academics from around the world working in the fields of machine learning and artificial intelligence examined computer vision in the context of BLOCK WORLD. (Sewak, M.2018)

1970s-80s:

The "theory of computation," which David Marr put out as a theory of visual processing in the 1980s, had a significant influence on the advancement of computer vision. According to David Marr, visual processing can be divided into three levels: the computational level, the algorithmic level, and the physical implementation level. The computational level is a high-level description of the processing tasks of the human visual system, the algorithmic level describes the intermediate steps of visual processing tasks, and the physical implementation level focuses on the neural mechanisms and neuronal activity required to implement these algorithms. The "Theory of Computation" by David Marr offers a framework for deriving explanations for and forecasts of the behaviours of vision processing. His work offers direction for computer vision theory research and serves as a crucial resource for dealing with real-world vision processing issues. (Marr, D.2010)

1980s-90s:

In 1989, Yann LeCun developed a network for identifying handwritten digits, building on the work of Japanese scientist Kunihiko Fukushima, who developed the neocognitron, a crucial neural network for image recognition. Previously known as LeNet, CNN was given his name after the person who created it. The LeNet-5, which outlines the fundamental elements of CNN, can be thought as the invention's foundation, however due to the scarcity of technology, particularly GPUs (graphics processing units), there was little study on CNN between 1990 and 2000. (Bhatt, D. et al.2021)

2012:

The success of AlexNet in 2012 opened the door for computer vision applications, and since then, numerous different types of CNNs, such the R-CNN series, have been developed. CNN models currently drastically diverge from LeNet while being all based on it. Also because of the success of AlexNet, the image classification potential of CNN is well known from that time, which means the rise of CNN algorithms. (Bhatt, D. et al.2021)

Current:

The science of computer vision has advanced quickly since the wave of artificial intelligence arrived. As an illustration, let's look at target identification. In 2012, AlexNet made a big impression at ImageNet, and in the years that followed, a number of models, including VGG,

GoogleNet, and ResNet, also broke records in the ImageNet competition. Thereafter, a series of deep learning-based target detection models, including R-CNN, Fast R-CNN, Faster R-CNN, and Mask R-CNN, were released, increasing the efficiency and precision of target detection. The YOLO family of models has recently drawn a lot of interest as well, with YOLOv4 earning a mAP of 43.5% on the COCO dataset. Moreover, the range of computer vision applications is expanding, including autonomous driving, facial recognition, and medical image processing, and they are now a crucial component of artificial intelligence technology. It can be mentioned that the field of computer vision is undergoing quick changes and constant evolution, and the recent developments have produced impressive results.

2.2.2. Present situation

Medical field:

The brain tumours with the highest rates of mortality and prevalence are gliomas. For treatment planning and post-treatment evaluation, accurate segmentation of gliomas and their intratumoral features is crucial. Unfortunately, manual segmentation takes a long time and is prone to difficult-to-define inter- and intra-rater mistakes. The majority of techniques for segmenting brain tumours rely on probabilistic models to forecast segmentation. Currently, CNN counting is used to segment tumour images, considerably increasing the segmentation effectiveness. (Pereira, S.2016)

In comparison to all other cancer types combined, skin cancer diagnoses are at an all-time high due to the rapid rise in global air pollution and ozone layer depletion. Machine vision research has now demonstrated its ability to offer cutting-edge skin cancer detection methods that are very accurate and can be used early in the disease. Time is the most valuable resource in the world for patients. (Javaid, A.2021)

Manufacturing:

One of the main technologies for achieving intelligent production is machine vision, which has successfully replaced manual visual inspection. A machine vision system uses optical devices and non-contact sensors to autonomously capture and analyse images of actual objects. One of the highest forms of human perception is vision. Our perception is greatly influenced by images. Human perception, however, is restricted to the electromagnetic spectrum's visible region. From gamma rays to radio waves, machine vision inspection technology can handle the whole electromagnetic spectrum. Machine vision can complete many jobs that human vision cannot by using robust vision sensors, carefully engineered light transmission techniques, and image processing algorithms. (Ren, Z. 2021)

Employers must take action to safeguard the safety of workers in the industrial industry. Yet, because it is challenging for selected observers to efficiently watch numerous screens at once, manual process monitoring is frequently ineffective. A safety problem can have a significant effect on both the personnel and the industrial organisation. With the aid of computer vision technology, personnel can efficiently identify problems with safety precautions and generate reports and alerts. In the event of an issue, the system may also automatically generate alerts, enabling management to take the appropriate steps right away to protect employee safety.

Hence, utilising computer vision technology is essential for businesses engaged in manufacturing. (Guo, B.H.W.2021)

Agriculture:

The tension between global population expansion and food shortage can be greatly helped by plant factories. These automated production processes heavily rely on machine vision technologies. Almost every phase of production is covered by the technology. Color, size, texture, and other phenotypic characteristics of plants reflect their state of health and growth. In the past, this data could only be gathered using empirical measuring techniques or visual observation, which called for a high level of grower knowledge and rendered real-time data collection unfeasible. Machine vision has been mentioned in many papers as a solution to these problems. High humidity and temperature in plant factories create a conducive environment for pests and illnesses to thrive, which can result in irreparable financial losses if not controlled in a timely manner. Pests and illnesses in conventional greenhouses can only be managed manually through observation and alarms, which is not only time-consuming and labor-intensive but also vulnerable to human subjectivity. Machine vision, a new technology that can quickly identify pests and illnesses and take the required precautions to ensure effective and stable agricultural production, is now being used in research to monitor plant health levels.(Tian, Z.2022)

Automotive:

The fast advancement of CNN technology in recent years has led to the maturation and incorporation of vision-based driver aid systems into automobiles. For instance, Tesla and the famous BBA (BMW, Mercedes-Benz, and Audi) are both working on this technology. Cameras take more work than other sensors to fully utilise their capabilities. Machine vision measures provide information regarding physical quantities that are utilised as benchmarks for vehicle control, such as distance and/or speed. To extract information that may be used for vehicle control, the different characteristics and patterns of the camera necessitate a thorough processing technique comprising calculations and algorithms. Yet, commercial driving assistance systems have long made extensive use of machine vision. Of course, there are some issues with the technology that are now being debated, including the abuse of the function by drivers that results in risky driving and the potential for some algorithms to contain flaws that might cause traffic accidents. Such incidents are, however, extremely uncommon, and as legislation and auto production advance, this technology will advance and become more useful to people.

2.3. Design and develop

The research will explore with the MOT20 dataset and employ two open-source programs, YOLOv5 and DeepSort, as detectors and trackers for object detection. To increase computing performance, the Python programming language will be the primary tool, together with NVIDIA drivers and hardware. Torch will also be utilized as the project's primary tool, with

Anaconda serving as a tool for testing and configuration. PyCharm will be utilized as the code editor for authoring and debugging code. The project will carry out several tests and data analysis to ascertain the accuracy of the monitoring utilizing these methods and datasets. With the goal of the project, more consideration must be given to elements that might influence the accuracy of the monitoring, such as the deployment angle of the cameras and the dataset's data, as well as the weighting files after training.

2.4. LSEPI considering

In both project design and development, as well as in writing articles, it is essential to pay attention to copyright issues. When using open-source code, it is necessary to check whether the code's source and the publisher hold licenses, as some publishers may not have proper accreditation. Using such data may lead to piracy issues. If using other databases or source codes, it is necessary to provide written explanations. When creating articles, it is crucial to promptly cite the sources that support the viewpoints or examples presented. Additionally, it is crucial to consider data privacy and confidentiality to ensure that the data used complies with relevant legal provisions, such as GDPR and HIPAA. As for social and ethical issues, it is necessary to consider discrimination, privacy infringement, and whether the project could cause social panic. In terms of professionalism, it is necessary to consider the reasonableness of the development and research methods, the accuracy of data and algorithms, and the results' interpretability and replicability. Besides these aspects, it is necessary to consider whether the project's application is legal. A classic example is the Deepface project, which was initially used for face recognition. However, some unfriendly individuals used DeepFace to swap faces in pornographic videos, thereby harming others' related rights, which is illegal.

2.5. Literature conclusion

The science of computer vision has existed for many years. Computer vision projects first appeared in the 1960s. Machine vision technology has steadily developed into one of the hotspots for research and application thanks to the continued advancement of computer technology and the expansion of applications. In order to acquire, process, analyse, and comprehend visual information, machine vision technology uses the intersection knowledge of several disciplines, such as computer science, pattern recognition, and image processing. This enables computers to perceive and cognize visual information similarly to humans. The application scenarios for machine vision technology are expanding and drawing attention because of the ongoing development of smart gadgets, autonomous vehicles, security monitoring, medical imaging, gaming engagement, and other sectors.

One of the key areas of machine vision technology applications is object statistics, which uses image analysis and processing to recognize, track, and count the objects in a picture. Several industries, including manufacturing, the courier business, the management sector, government agencies, and others, use object statistics technology. With object statistics technology, it is possible to manage objects in the environment in real-time while also increasing work

efficiency and safety. As a result, object statistics technology has reached a cutting-edge stage. The computer vision sector will experience bigger breakthroughs with such a development trend.

The most cutting-edge techniques and material were not used in this investigation since the stability of the programme had to be considered during design and development. The goal of this article is to provide the reader with some fascinating ideas and insights by thoroughly describing the technology in use and include presentations of related goods and design influences for the project.

Here are some predictions for future trends:

(1) New principles used in CNN's design have changed the focus of study, notably in the area of computer vision. Studying advancements in CNN's architecture is a promising research topic with the potential to become one of the most popular AI strategies.

(2) CNN is planning research in the field of ensemble learning. The model may enhance the generalization and robustness of numerous categories of pictures by integrating multiple and diverse designs by extracting distinct semantic representations.

(3) Attention is a crucial process in the human visual system for acquiring information from images. Furthermore, the attention mechanism collects the crucial information from the image and stores its context in relation to the other visual components. In the future, the research could be conducted to preserve the spatial importance of objects and their distinguishing characteristics during subsequent stages of learning.

(4) The human visual system's attention process is essential for extracting information from pictures. Moreover, the attention mechanism remembers the context of the image in respect to the other visual elements and gathers the essential information from the image. In the future, research might be done to maintain the spatial significance of objects and their distinctive qualities throughout succeeding learning phases; (Bhatt, D. 2021)

(5) Fusion of multiple sensor data: To produce more thorough and precise object statistics, future machine vision will combine many sensor data, including depth sensors, LiDAR, etc.

(6) Autonomous learning and self-adaptation: In the future, machine vision will pay more attention to autonomous learning and self-adaptation, allowing the algorithm to modify and improve itself in response to changing environmental conditions, thereby enhancing the accuracy and stability of object statistics.

3. Requirements analysis

initial article:

‘When a university building has to be evacuated, teaching staff has to report which room they were in and how many students were there and were evacuated. The idea for this project is that image recognition could be employed to count the number of people who are present in a classroom in real-time. When a building is evacuated it is possible to know how many people were present in each room. The system would know whether everyone evacuated and could provide an all clear for each monitored room.’

Claim:

Through study of the initial concept and introspection, the requirements are described below. Due to the fact that I was working alone and lacked the funds to buy the necessary testing and experimental tools as well as access to any of the school's CCTV, I was unable to accomplish this project by analysing the live CCTV feeds. The access to the live CCTV feeds also required other specialised concerns that I was not yet fully versed in. Furthermore, I didn't have enough time to study and resolve these problems because my delivery was scheduled for April 2023.

3.1. Functional and unfunctional requirements

3.1.1. Functional requirements

1. Image input:
An image or video input source is required, but due to development time and team size constraints, it is not possible to address how to use the live surveillance feed.
2. Object detection:
A program with image processing and recognition capabilities, such as YOLO or R-CNN, is needed for real-time detection of whether someone is in the classroom.
3. Object tracking:
A program with people tracking and recognition capabilities, such as DeepSort, is needed to track the number of people in a classroom in real time.
4. Headcount:
To gather and process data from sensors and cameras in order to count the number of persons in each room in real time, a specialized real-time data processing system is needed.
5. Algorithms for determining behaviours:
To count the individuals in each room, a headcount estimation technique must be used in conjunction with an item identification and tracking system.
6. Dataset:
A relevant dataset containing reliable information is needed. One of the main

contributing factors to the model's correctness is the dataset, which must undergo extensive testing and validation to guarantee the system's dependability and stability.

7. Fire Alarm:

The system must be capable of spotting fires and sounding the proper alarms to warn people to flee. The system should be able to count the number of persons still inside each unattended room in the case of a fire.

8. Safe evacuation:

In order to ensure that people may leave the building swiftly and safely, the system must be able to provide instructions for the safe evacuation of each room.

9. Automatic processes:

Object detection and object tracking need the usage of two separate computer algorithms in tandem. To create the most ideal system, the best variants, including R-CNN, mask-RCNN, YOLO, and other YOLOv5 iterations, should be tested and used.

3.1.2. Unfunctional requirements

1. security:

This system must ensure the security and confidentiality of the data by ensuring that only authorized employees is able to access or modify the data. The goal of the system's security is to safeguard user privacy, ensure the integrity and confidentiality of data, prevent unauthorized access and abusive use, and minimize losses or damages resulting from security flaws or attacks.

2. User-friendliness:

In order for the faculty and staff to quickly respond to emergencies and accurately provide information and advice, the system must be simple to use and understand. This means that the system needs to have an easy-to-use interactive user interface. Because of this, nobody will be able to understand the code or know how to use it. When an unexpected issue arises, asking a technician to remove and analyze the system is not realistic.

3. Extensibility:

This system must be adaptable so that the surveillance may be easily expanded, or the processing capacity can be increased as needed, without compromising the system's reliability or performance. It is important to consider the project's potential for future expansion. For instance, adding a function to identify ignition points, analyzing the best escape routes, or even connecting the programme to the Web to develop a special application for firefighting services.

3.2. Moscow analysis requirements

Function requirements	
MUST	Image input: The necessary image input is a basic requirement for this item, if there is no image input then there is no detection source. Object detection: Object detection is one of the essential elements to achieve the headcount function. Object Tracking: Object tracking is one of the essential elements to achieve the headcount function. Headcount: This is a mandatory feature for this project, if this feature is not implemented then the project will not be completed. Behaviours determination algorithm: Once the headcount has been implemented then a determination needs to be made as to which direction different people are walking in order to know if someone is entering or exiting. Without this feature the room count would not be real time.
SHOULD	Data set: Given that this project is for fire evacuation, the application will eventually be deployed in a university or public room. This would mean that the program would receive CCTV-like video. The angle of the video will be on the ceiling or in a corner. So, a suitable data set will affect the accuracy of the program. So, this is a functional requirement that should be in place. Automated procedures: Because the development of the project involves different parts of the code, but in the end, they should be integrated together to form a more automated program.
COULD	Safety evacuation: The function of safety evacuation is optional for this program, as it would be more

	appropriate to do this task manually. For example, in the event of a fire, firefighters would use the system and organize the evacuation. Although this function could be extended in the future, when the program is more mature, it is not necessary at present.
WOULDN'T	Fire alarm: This function would be less effective than the safety evacuation function, firstly when a fire alarm is triggered. The existing building is already equipped with an alarm, so there is no need for this design to be redundant.

Unfunction requirements	
MUST	Usefulness: Since the primary goal of this project is to aid in fire evacuation, and since it is based on the idea that machine vision can assist in fire evacuation, there is a clear limitation to this project. Since surveillance probes will be used to gather the headcount, this programme will probably be used in the surveillance room. Hence, most of the monitoring rooms are situated next to the security room. Security staff will be the system's initial user, but they won't be able to understand its design principles, thus there needs to be an interactive interface that is simple to use and observe.
SHOULD	Security: As mentioned in the ease of use, this design will eventually be deployed within the security room monitoring system, so it is inherently secure. However, care will need to be taken with regard to the management of operators and whether the use with the surveillance system will result in security breaches, but at present, due to time and equipment resources, this design will simply add a login interface. Security issues will be expanded in the future.
COULD	Scalability: Considering I am the sole designer, developer, and tester for this project. So, the functionality

	of this project won't be further expanded. Further components will be added in the future if more personnel and resources are made available to aid in the creation of this project. The design expansion could only be what it could have been at this stage.
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4. LSEPI problem

4.1. Legal

The issue of CCTV laws cannot be avoided, as the project's features necessitate the installation of cameras. Since the government's deployment of Surveillance began, there has been considerable opposition. Midway through the 1990s, when CCTV systems began to be installed in the urban areas of England and Wales, all local attorneys stated that under British law, such surveillance was entirely superfluous. They pointed out that there is no right to privacy in this country (at least not one that is independent of property rights), so individuals who are dissatisfied with being under surveillance have little to fight against. This altered in 1998, when the Human Rights Act of 1998 was passed. The Act stipulates that individuals have a right to privacy, even in public areas. (Gras, M.L.2002)

Obviously, as the applicable laws improve, this legal issue becomes less and less problematic. In addition to the use of cameras, this procedure must also account for the collection, processing, and storage of data. Since the passage of the Data Protection Act of 1998, the use of cameras in public places is governed by its provisions. The law requires that data be collected, processed, and stored in conformance with the principles of data protection. The Copyright, Designs, and Patents Act of 1988 is also an issue, as other documentation and open-source documents will inevitably be used in the design and development of this endeavour. Therefore, it is essential to pay greater attention to copyright issues and verify that open-source files are permitted and licenced prior to their use. Whenever required, references to literature must be indicated in a punctual manner.

1. Data Protection Act 1998: The use of cameras in public places must comply with the provisions of the Data Protection Act 1998. The law states that personal data must be collected, processed, and stored in accordance with the data protection principles.
2. Human Rights Act 1998: The right to privacy is protected by the Human Rights Act 1998. This means that if an individual is recorded, they must be informed and have the right to object.
3. Copyright, Designs and Patents Act 1988: It gives creators of literary, dramatic, musical, and artistic works the right to control how their material is used. Rights include broadcasting and public performance, reproduction, adaptation, distribution, rental, and lending of copies to the public.

4.2. Social

It is important to consider several elements, including human rights, digital media, and technology, to make sure that this initiative will have a good influence on society. A crisis of trust between the public, the government, or the police may result from the usage of cameras in public spaces, according to studies. This will increase mistrust and rifts among the neighbourhood. The use of cameras in public locations sometimes results in inequity and

prejudice. For instance, if the camera only observes a few groups, this might result in an unjust treatment of those groups. The project must consider any hazards that data gathering, and utilisation might have. Trust in the company responsible for data collection might be seriously damaged by the abuse or poor treatment of information. Thus, it is crucial to implement the necessary safeguards to preserve individuals' privacy and guarantee the avoidance of any unlawful access to or use of personal data. The project should consider any possible environmental effects of pollution, particularly if it will use a lot of energy. The project's sustainability and any possible environmental costs connected with its execution must now be considered. (Gregory, S. 2019)

The larger influence on society, such as its social and economic effects, should also be considered. It is important to ensure that the project's advantages are evenly distributed and do not have any unforeseen side effects.

4.3. Ethics

Ethical developers must consider a project's legal, ethical, and social impacts in addition to its technical ones. The target user's surroundings must be considered while developing and using technology. Before commencing any project, it's vital to thoroughly analyse possible applications and their impact on the individual and society. This involves assessing the risk of technological abuse and unanticipated consequences.

Deepfake technology in pornography raises legal, social, and ethical issues, thus it must be used ethically. Face recognition technology in public areas raises privacy and surveillance concerns that might stifle innovation and vibrancy.

Ethical developers must consider how projects will affect society and the ethical and moral issues that come with the technology they produce. This means considering the impacts on people, communities, and the environment and being as open and honest as possible about development programmes and individual rights and freedoms. Speak out against technology abuse and promote ethical norms to ensure technology is created and used responsibly and ethically. (IEEE, directorate. 2020)

4.4. Professional

The developing must follow professional standards while designing technology, including using public cameras and collecting, processing, and storing personal data. The developing must follow ethical standards that encourage openness, accountability, and respect for individual rights and freedoms.

The obligations of public tech users must also be considered. They must handle personal data professionally and legally. This involves protecting data from unwanted access and ensuring that public surveillance cameras respect privacy and prevent oversurveillance.

Technology development requires code and model security and truthfulness. This needs ongoing testing and validation to find and correct vulnerabilities that might undermine the technology's security or accuracy.

I must assure ethical and responsible usage of my technology as a developer. This requires continual education and professional development to keep up with industry advances and create and follow ethical, legal, and professional standards. So, I can assure that my technology benefits society without causing harm.

4.5. GDPR

In order to comply with the GDPR, it is necessary to thoroughly consider a number of factors. First, the legality of data processing must be established. Therefore, the collection of data must be lawful and reasonable, such as by specifying in the contract the obligations that both A and B must fulfil and for which they are legally liable. Before collecting and processing individuals' personal data, organizations must also obtain their explicit consent and inform them of their rights under the GDPR (Paul & Axel, 2018).

Additionally, effective technologies and procedures are required to ensure the security of personal data (Custers, 2018). This includes identity obscuring, the implementation of security policies, and routine backups. In addition, it is essential to appoint a Data Protection Officer (DPO) who will comply with GDPR regulations and handle any data protection issues that may arise (Custers, 2018).

Lastly, a mechanism for responding to requests from data subjects, such as requests for access to personal data or requests for data deletion, must be in place (Custers, 2018). And when a data compromise occurs, the incident must be reported to the appropriate government agency within 72 hours. (2018). (Custers, 2018).

5. Similar product analysis

5.1. Fundamental Algorithms

CNN, RNN, and GAN are the three most prevalent computer vision algorithms in use today. This project focused on selecting the most suitable framework among CNN, RNN, GAN algorithms, as the data and literature review revealed that CNN algorithm is the most suitable for this project and most CNN-based algorithmic frameworks are open-source. It used by a sizable number of people, and it was able to identify many development communities and resources that needed to be utilized.

CNNs are typically employed for duties such as image and video processing. CNNs can autonomously acquire image features and be used for classification, segmentation, detection, etc., because their convolutional layer allows for the extraction of image features and multi-layer convolution and pooling operations. CNNs offer 1. (RCNN, Mask RCNN, YOLO, LeNet, AlexNet, VGG, Google Net).

RNN (Recurrent Neural Network): Primarily utilized for processing data with sequences, such as natural language, audio signals, and data such as time series. Due to back-propagation algorithms and long short-term memory (LSTM) units, RNNs can model sequential and regular information in sequential data, facilitating speech recognition, machine translation, and language generation.

RNN (Recurrent Neural Network): It is used primarily for processing sequences of data, such as natural language, audio signals, and time series. Due to the back propagation algorithm and the Long Short-Term Memory (LSTM) unit, RNNs can model sequential and regular information in sequential data, allowing them to perform tasks such as speech recognition, machine translation, and language generation.

GAN (Generative Adversarial Network): This algorithm is a model of unsupervised learning used to generate and create data such as images, audio, and video. GAN is comprised of two essential components: the generator, which is responsible for generating false data, and the discriminator, which is primarily responsible for identifying and contrasting real data with fake data generated by the generator. The GAN can autonomously learn to generate distributions of real data and generate artificial data that resembles the real data using adversarial training. The GAN is primarily utilized for duties such as image generation, image restoration, and image conversion.

5.2. Existing open-source algorithms

The more mainstream open-source algorithms are currently YOLO and Mask RCNN, and YOLO is the algorithm that has been decided to be used for this project.

YOLO v5 is a deep learning-based target detection algorithm, it is not the latest version in the YOLO family. The latest version is YOLO v8. v5 has a lightweight network structure with impressive detection speed and accuracy. v5 has at its core a single-stage detector based on a backbone network, which detects targets by using the Anchor-Free method. Furthermore, YOLO v5 supports multi-scale training and testing, so it can perform target detection on images of different resolutions. This is why YOLO v5 is so versatile.

Mask R-CNN is a deep learning-based instance segmentation algorithm, and one of the most recent additions to the R-CNN family of algorithms, which evolved from Fast R-CNN to Faster R-CNN. A region extraction network for target detection and a Mask network for pixel-level segmentation make up the core of the Mask R-CNN framework. By jointly training the two sub-networks, the Mask R-CNN algorithm obtains outstanding performance on both target detection and pixel-level segmentation tasks. (He, K.2018)

YOLO v5 is fast and can achieve real-time target detection, and YOLO v5 has a high degree of real-time performance. However, Mask R-CNN is relatively slow and requires more computational resources and a lot of time to complete the detection task. Because Mask R-CNN is primarily used for object instance segmentation, it takes detection a step further by segmenting the interior of the object so that each object detected can be segmented accurately. Although the accuracy of YOLO is not as good as that of Mask R-CNN, the accuracy of YOLOv5 is sufficient for this project, and YOLO v5 uses some new techniques and improvements, such as adaptive convolution and PANet, which improve the detection accuracy. Therefore, YOLO v5 is more suitable for applications that require real-time target detection, such as autonomous driving, UAVs, and surveillance footage, among others. Mask R-CNN is more suitable for scenarios that require object segmentation and more accurate target detection, such as medical imaging.

5.3. Similar product

A survey of existing products revealed a few products that presently exist and are relatively well-suited to the project's design concepts. There are also products for directing pedestrian movement, detecting vehicle flow, and measuring factory footfall. These items served as a source of inspiration for the project's design. One product for pedestrian direction tracking implements the function of pedestrian movement direction but is suitable for non-enclosed environments and simply monitors the direction of human movement without regard to the number of people present or whether the last person is still in the area; the same is true for the product for vehicle flow monitoring. Another product is used for footfall monitoring in factories,

but it is only concerned with the total number of people in the monitored area and does not determine whether people are entering or departing. After studying and analysing these similar products, the project adopted what was best from them and optimized the design along their lines.

6. Design

6.1. Confirm Objectives and Literature Reading

Before beginning the process of designing and researching the project, it is essential to conduct a thorough analysis of the initial project requirements. This will guarantee that the project does not stray from its intended purposes and goals.

These are the original requirements that were stipulated:

" When a university building has to be evacuated, teaching staff has to report which room they were in and how many students were there and were evacuated. The idea for this project is that image recognition could be employed to count the number of people who are present in a classroom in real-time. When a building is evacuated it is possible to know how many people were present in each room. The system would know whether everyone evacuated and could provide an all clear for each monitored room."

The first sentence of this paragraph reveals that the system would be utilised indoors, and more specifically, in classrooms at schools. This is one of the most important elements that we can conclude from this paragraph. The following sentence makes it quite evident that image recognition is required for the project in order for it to accomplish its goals, and the third sentence makes it abundantly clear that the system requires a headcount function. At the very last sentence, it is made quite apparent that there must also be an alert function. It is essential to take note of the fact that the alert feature will not be put into place because there is no requirement for it. This is additionally discussed in the part devoted to the requirements analysis. As soon as the goals have been articulated in detail, it is time to investigate the technologies that will be necessary for the project. At this stage, it is possible to zero in on a select few key terms, including "indoor," "machine vision," and "headcount." Reading books that contain or relate to these three topics is a good place to concentrate one's efforts in order to finish the job more successfully.

6.2. Design ideas

The design of this project was inspired by the investigation and study of other items that are similar to it. During the course of the research conducted on pedestrian and vehicle traffic detection products, it was discovered that the majority of the products have only one determination area. The principle behind these products is to count the traffic by calling DeepSort's tracking function to determine whether the centre point of the object collides with the determination area, and to determine the direction based on the direction of movement of the centre point. This was discovered during the research conducted on the products. In addition, these are placed in an open outdoor area at a predetermined location. Yet, the environment that

one finds inside a building is not the same as the environment that one finds outside. Not only does the design add one decision zone to two decision zones, but it also adds one decision zone to better discern whether a pedestrian is entering or exiting a room. This is done so that the tracking accuracy can be improved. It is not necessary for the design to monitor the entirety of the space; rather, it is sufficient for it to watch and count all of the room's entrances and exits. A fixed determination area would result in the determination area being off the entrance or exit, which would make the system difficult to use if the code had to be manually debugged each time it was deployed. This was another thing that was discovered during the preliminary experiments and tests. Because the architecture of each room was unique, a fixed determination area would result in this. As a result, the system was enhanced to allow for a more expedient selection of the decision area.

6.3. System framework design

Anaconda can help make it easier and faster to manage and deploy dependency sources, and it is distributed using the Python and R programming languages, both of which are available under an open-source licence. Machine learning, the processing of large amounts of data, and predictive analytics are just some of the applications that can benefit from using Anaconda. programme as well as any dependencies necessary for file management. (Conda, documentation.2017)

Python 3.9 is the computer language that is being used for the project, and PyCharm is being utilised as the integrated development environment for the code. It is essential to employ a hardware graphics card for acceleration purposes given that the project focuses on the identification of video streams rather than a single image. torch, which is a machine learning library, is a necessary dependency for the project in order to make use of GPU acceleration.

Both YOLOv5 version 3.0 and DeepSort were utilised in order to recognise and track moving objects. In terms of the detector, YOLOv5 version 3.0 is a more stable version that yet retains some performance. These statistics on stability and performance were gathered on the internet as well as through my own personal testing in a limited space. The creators of YOLOv5 have not published a paper on the topic, thus these are the only sources available. When tracking a target, tracking can be accomplished by predicting the trajectory of the target's position in a continuous image. This allows for more accurate tracking. This is accomplished by making an educated guess as to the path that the object will take in each frame.

A straightforward sign-in screen and an operator interface are going to be included in the capabilities of this user interface. This interactive interface will contain a capability to quickly select a zone as well as a function to count the number of attendees. This user interface (UI) is functionally equal to invoking YOLO and DeepSort, which enables the project to achieve its intended goal.

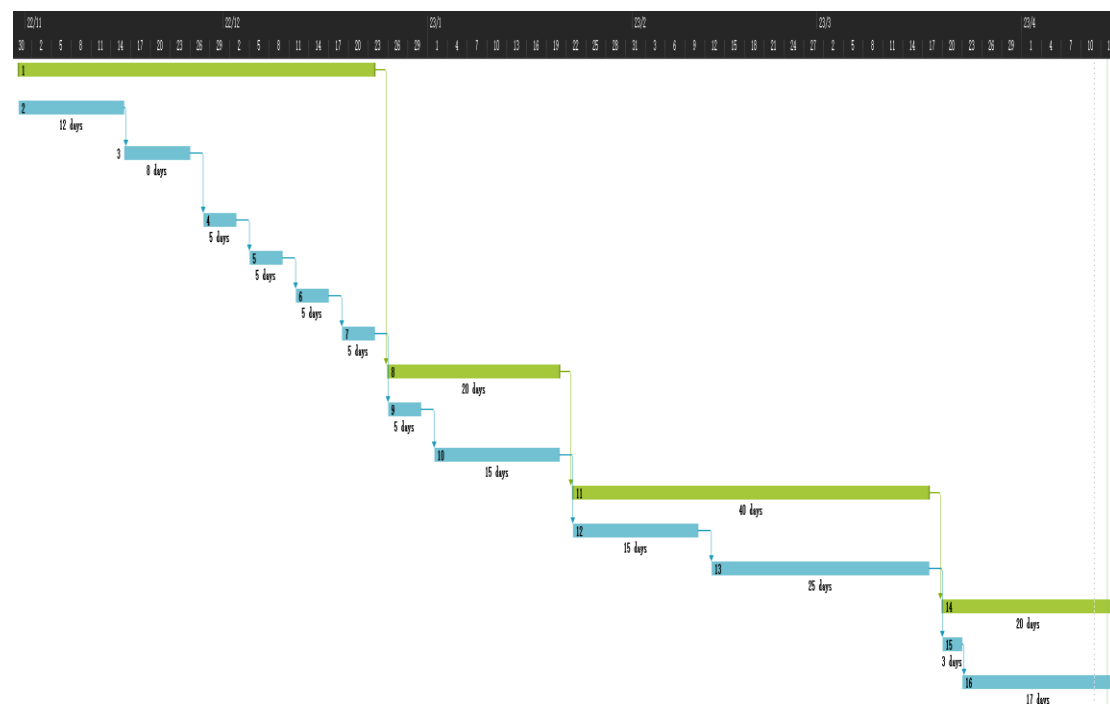
6.4. Based on incremental development model developing

/	AUTO/MANUAL	NAME	Duration	From	To	Priority
1	✈️	<i>Early Period</i>	40 days	2022/10/31	2022/12/23	
2	📅	Comparing algorithm performance	12 days	2022/10/31	2022/11/15	
3	📅	Setting up the environment and installing dependencies and drivers	8 days	2022/11/16	2022/11/25	2
4	📅	Collecting datasets	5 days	2022/11/28	2022/12/2	3
5	📅	Training	5 days	2022/12/5	2022/12/9	4
6	📅	Testing experiments	5 days	2022/12/12	2022/12/16	5
7	📅	Confidence analysis	5 days	2022/12/19	2022/12/23	6
8	✈️	<i>Mid-term</i>	20 days	2022/12/26	2023/1/20	1
9	📅	Attempting to improve confidence levels	5 days	2022/12/26	2022/12/30	7
10	📅	Test analysis	15 days	2023/1/2	2023/1/20	9
11	✈️	<i>Late stage</i>	40 days	2023/1/23	2023/3/17	8
12	📅	Combine with DeepSort tracking	15 days	2023/1/23	2023/2/10	10
13	📅	Called and counted for coding	25 days	2023/2/13	2023/3/17	12
14	✈️	<i>Final</i>	20 days	2023/3/20	2023/4/14	11
15	📅	Optimize parameter passing and counts	3 days	2023/3/20	2023/3/22	13
16	📅	Designing the interaction interface	17 days	2023/3/23	2023/4/14	15

Working breakdown structure

Green = stage

Blue = every single stage task



There are certain distinctions between the incremental model development method and the waterfall model, despite the fact that the incremental model development method is a variant on the waterfall model. The waterfall model is a well-known type of software development model. It consists of several phases, including requirements analysis, design, coding, testing, and operation and maintenance. Each step moves in a sequential order, and all phases need to

be finished before going on to the next phase. Under the incremental approach, the software creation procedure can be broken down into a number of main phases, and each major phase can be further subdivided into a number of minor phases. At the end of each significant phase, a particular function is finished, and it is then included into the system. The incremental model is not merely a redistribution and iteration of the waterfall model; rather, the most essential component of the incremental model is the feedback obtained at each iteration, which enables a more adaptable approach of software development.

The project also benefited from the incremental development model because there were always unanticipated challenges that developed during the development process. In order to recap and summarise the issues that were experienced each time a significant phase was completed, the project benefited from the incremental development model. There are a total of twelve lesser phases within the project's four major stages. There is a total of six small stage in the start-stage, two small stages in the mid-stage, post-stage and final. This is made clear in the diagram that is located above as well as in chapter 7.

6.5. Summary and Outlook

The requirement to acquire and use expertise in a variety of fields, such as machine learning, computer vision, software development, and testing, is the main cause of the difficulty presented by this project. In particular, expertise in image processing, target recognition, and deep learning, as well as the appropriate techniques and tools, such as OpenCV and TensorFlow, is necessary. In addition to this, it is essential to develop and implement a headcount determination function that is both accurate and efficient, as well as to think about how to make the programme user-friendly and simply accessible. During the course of the development process, it was necessary to make ongoing trade-offs, conduct testing, and engage in analysis in order to guarantee the consistency and dependability of the programme. The initiative is valuable to society because of its benefits, despite the many difficulties that must be overcome. If it can be developed effectively, it will have a good impact on the safety of people's lives as well as the protection of their property when it comes to fire. Therefore, it makes perfect sense to select this project as my final year project, and this is excited about the prospect of learning new things and increasing my skill set while working on its growth.

7. Developing

7.1. First stage

7.1.1. Overview of Algorithm Performance

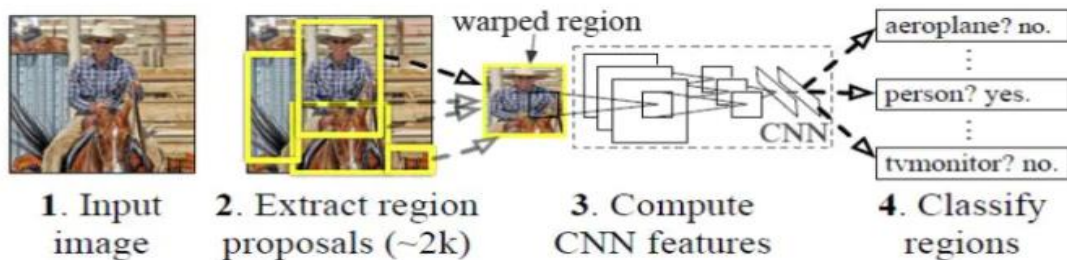
Before 2012, target detection algorithms mainly used manual feature extraction as a means of target recognition, which is less calculation intensive. the RCNN algorithm was released in 2014.

The RCNN algorithm is generally divided into four steps:

- (1) segmentation of the image into more than one thousand candidate regions.
- (2) Extracting the features of the candidate regions by means of a deep learning network.
- (3) Inputting the previously extracted features into a support vector machine for classification.
- (4) Further modification of the selection frame.

The second stage of the deep network can be divided into two levels:

- (1) Using a selective search algorithm, a candidate wraparound box that can accommodate the target is proposed.
- (2) capturing these candidate envelope boxes for classification in the CNN algorithm. Figure 2.1 shows the steps of the RCNN algorithm.

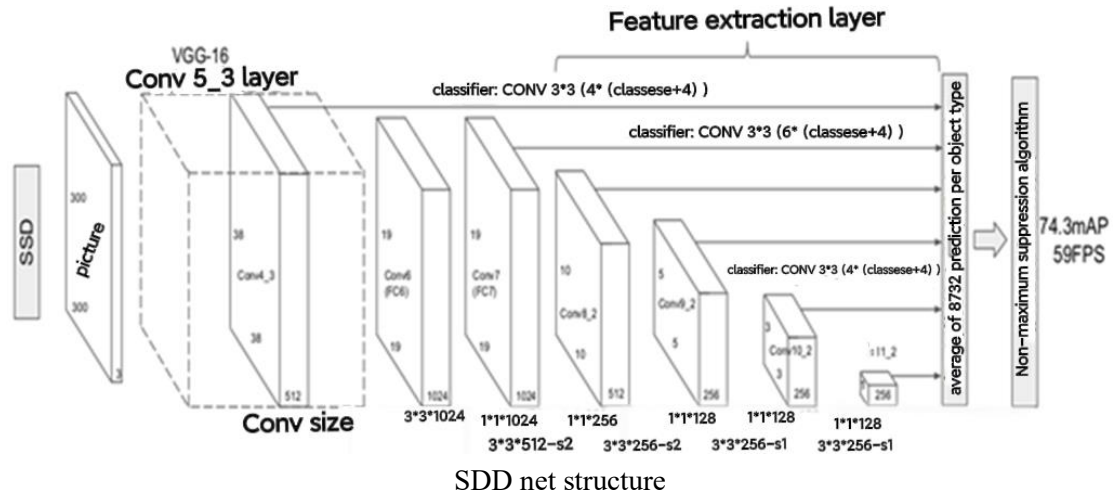


Steps of the RCNN algorithm

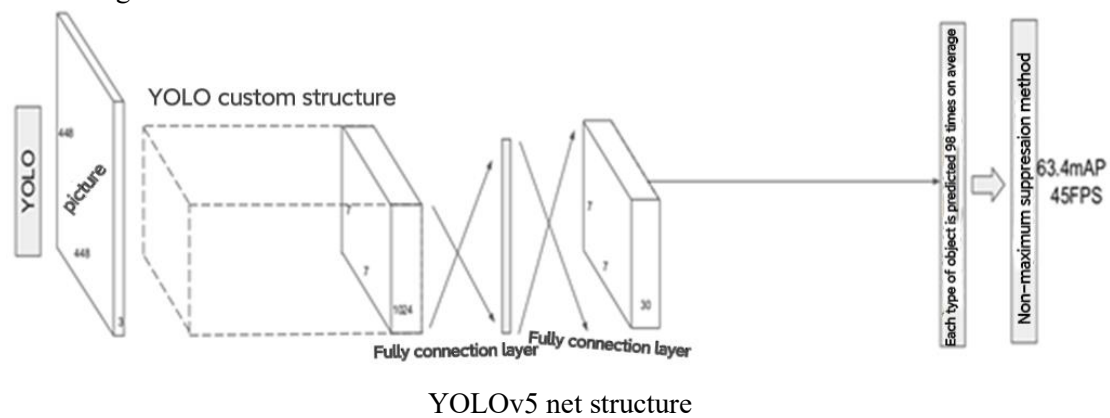
In late 2014, an improved RCNN algorithm, the fast RCNN algorithm, improved the accuracy and speed of the operation by reducing the time of the direct transfer process. The RCNN algorithm 2015 is faster and is a truly leading deep learning target detection algorithm, which is based on convolutional region target networks to implement envelope box targets, and gives confidence levels, and then forwards the RPN to the RCNN module for classification, where selective search is more necessary, greatly saving computation time and being more than 10 times faster than the RCNN algorithm. The common feature of this series of RCNN algorithms is that they all adopt a two-step strategy and have high accuracy of detection results, but the simulation speed is slow and hardly exceeds 6fps.

At the end of 2015, two typical detection algorithms based on embedded convolutional

networks were proposed. the SSD algorithm did not generate recommendations at runtime. Using VGG16 as the main structure, the feature mapping of each convolutional layer was integrated, replacing the last 2 fully connected layers with convolutional layers, and adding 4 convolutional layers. The outputs of the five different convolutional layers were convolved by two different 3×3 convolutional kernels.

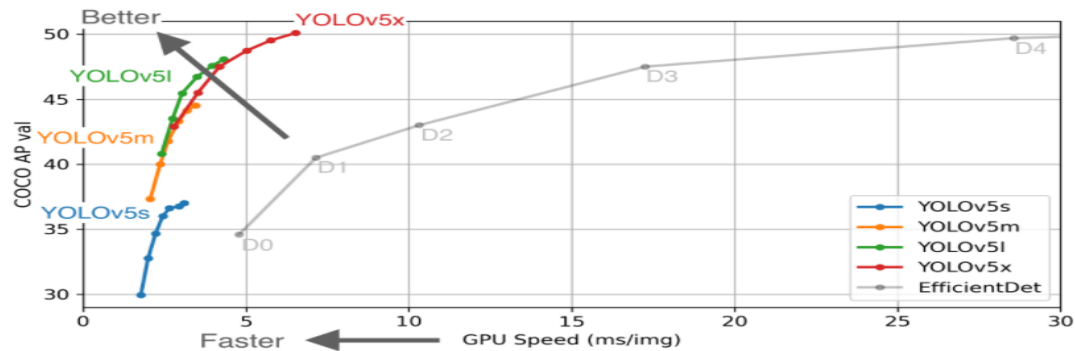


In the same year, the YOLO algorithm was introduced. The basic idea is to sample the image uniformly at various locations and use the CNN to extract features directly for classification and regression. a schematic of the network structure of YOLOV5 is shown in the figure. The YOLO algorithm will be described in detail later.



Yolov5 is the series version of the YOLO algorithms released by Glen Jocher on 25 June 2020. It has comparable performance to Yolov4, but at speeds of up to 140 frames per second. The model's small size and low deployment costs facilitate rapid deployment of the model. Unlike the previous four generations of YOLO algorithms, it is no longer based on the dark web framework, but uses the Pythoch framework. The framework is very user-friendly, making it easier for users to form their own datasets and put them into production. Not only can it recognise images individually or in batches, but it can also recognise video input from cameras in real time. PyTorch driver weight files can be converted to ONXX for Android, OpenCV or iOS via CoreML.

YOLOV5 has four network models corresponding to four different sizes of weight files, from small to large, YOLOV5S, YOLOV5M, YOLOV5L and YOLOV5X. The figure below shows the performance test deviations of the various YOLOV5 network model algorithms. The YOLOV5S array is the smallest, fastest and has the lowest AP accuracy, while the YOLOV5X array is the largest, slowest and has the highest AP accuracy.



YOLOV5 Performance testing of various network model algorithms

YOLOV5s has a very small network structure and weight file. The network structure of the original YOLOV5 is shown in Figure 6, where we can see that the YOLOV5 network consists of three parts: the backbone network, the head network and the detection layer, respectively. As an example, the input image size is 640*640, and the exact composition of the three components is indicated in the figure, as well as the number and size of the feature maps obtained after multiple convolutions of the original image.

The backbone network is composed of layers 0 to 9, alternating between a convolutional layer (Conv) and a cross-stage local network layer (C3), with two types of C3 structures, 3-fold and 9-fold, designed to address the problem of computational overload in inference from the perspective of network structure. Since the Focus layer has been removed from the latest YOLOV5 algorithm, we will not discuss it here. The addition of the Spatial Pyramid Pooling (SPP) layer in the eighth layer largely increases the size of the perceptual field, facilitating the detection of large targets and enhancing the non-linear representation of the backbone network.

The head network is 10 to 23 layers, with a positive pyramidal structure of FPN and an inverted pyramidal structure of PAN, consisting of 4 convolutional layers, 4 C3 layers, 2 upsampling layers (Upsample) and 4 splicing layers (Concat). The convolutional layers have 1*1 and 3*3 kernels, the C3 layers are 3-fold laminated and the upsampling layers are 2-fold upsampled. The fusion of feature maps can be enhanced by the combination of FPN and PAN, but there is



original YOLOV5 net structure

still room for improvement. The detection layer uses GIoU_Loss as the loss function, with three detection heads of sizes $20 \times 20 \times 255$, $40 \times 40 \times 255$ and $80 \times 80 \times 255$ respectively.

7.1.2. build up the environment and installing dependencies and drivers

First, check the existing hardware information of the computer, it is important to see if it has a separate graphics card. Here there are two commands to check the hardware information, the first one can view all information except the graphics card, the second one can display detailed information about the graphics card. win+r key combination type cmd to open the command control line. Enter systeminfo to see the hardware details and nvidia-smi (this command can only be used for NVIDIA cards) to see the detailed graphics card information. The appendix will contain details of the computer on which the project is being developed. Generally, the computer is equipped with a discrete graphics card by default and the manufacturer of the card is NVIDIA. then check the drivers for the card, as the Pytorch version number will correspond to the different CUDA and cuDNN versions. CUDA is a parallel computing platform and programming model invented by NVIDIA. There is also a necessary correspondence between CUDA and the graphics card driver version.(NVIDIA, Cuda.2022)

CuDNN is a GPU-accelerated library of deep neural network primitives. CuDNN provides highly tuned implementations of standard routines such as forward and reverse convolution, pooling, normalisation, and activation layers. It does not have a one-to-one relationship with CUDA. Before download cuDNN, it will also need to agree to Nvidia's ethical AI terminology. (NVIDIA, Cudnn.2023)

After installing CUDA and cuDNN, then need to install Anaconda. anaconda refers to an open-source Python distribution that contains more than 180 scientific packages such as Conda and Python and their dependencies. anaconda can help us to better install dependencies under different models, as we may have, we may install many different versions of Python, python, TensorFlow and other dependencies in different addresses of different models. Anaconda can therefore help us to better manage our dependency ring. If you wish, Anaconda even provides a GUI interface for the user to work with.

Once computer have installed Anaconda, it needs to create a new environment address and specify the Python version number of the environment (conda create -n FYP python==3.9). It should be noted here that YOLO was chosen as the solution for this project, but YOLO currently has several different versions which require different versions of Python. If the wrong version of Python is installed, the whole model will not work and some of the code packages in the different Python versions will not exist or will already refer to functions in other ways in updates and iterations. Of course, it's hard to find problems early in the configuration process, meaning that the project interpreter it needs to configure for the encoder current using may not be compatible with the Python it has installed in the environment. These different versions also caused the project to waste a lot of time on compatibility testing in the beginning. The project uses PyCharm as the IDE, which is a personal choice.

Once created, the environment requires the installation of Pythoch. PyTorch, a fully functional

framework for building deep learning models, a type of machine learning commonly used in applications such as image recognition and language processing. Written in Python, it is therefore relatively simple for most machine learning developers to learn and use. Here, it should note that the version of pythrch download needs to correspond to the version of CUDA computer have installed. (NVIDIA, pytorch.2023)

Next, pull the source code from GitHub. Here, selects the licensed code to pull, as that may be breaking the law without the author's permission. The reason for choosing to download the code directly rather than write your own is that if write the code directly from scratch, it is unlikely that the project will be completed unless it takes two or three years to write. (Jocher, Glenn. 2020)

It will then need to download the dependencies and open the anaconda command line and type: conda activate FYP to activate the previously built environment, then cd to the YOLO root directory and type the command pip install -r requirements.txt . Wait for the installation to complete.

7.1.3. Collect dataset

When the code is finally made accessible, it will not be easy to begin running and training immediately. The first thing that needs to be done is the creation of a dataset. Images make up the majority of the files in the primary dataset. Therefore, different technologies will have to be utilised in order to label the photos contained inside the dataset. The following stage is to acquire the pertinent image data, which is an essential step considering the fact that the quality of the dataset will directly affect the accuracy of the detector. There is a considerable possibility that an unclear weight file will be trained if the source of the photos, the angle at which they were taken, and the lighting circumstances are not consistent with one another.

As a developer, I began the process of training the model by searching the internet for different portraits and studying how they were composed. The results, which were, of course, to be expected, demonstrated that the model was not very precise and, on occasion, even mistook items of clothing for living beings. Therefore, in order to make dataset more consistent, I decided to use the data from the MOT. (Dendorfer, P. 2020)

The format of the dataset, which is analogous to the format of a dataset that has previously been annotated, was a new problem that occurred because of using the data from the MOTchallenge. The fact that this data format is not the same as YOLO or labelling, however, implies that the data themselves cannot be used; nonetheless, the format can be transformed. The dataset is converted using a piece of open-source software, but the converted data must be in the YOLO format for us to use it. The open-source code is only capable of converting MOT to VOC, so I proceeded to learn by myself on how to convert VOC to labelling and finally labelling to YOLO. (PanXF-HUST.2020)

7.1.4. Training

After the data has been labelled, it is necessary to start training the dataset. At this point, a YOLO weight file, called a pre-training weight file, needs to be downloaded. This file is available in different sizes. It is recommended that to choose as many as need (depending on the detection model, image size and computational power). After this step, the code needs to be configured, as a result of which a choice of input sources and output addresses as well as the location of the pre-trained weight model is required. Here a yaml file in the data and model folders in YOLO needs to be modified by this. First make a copy of the coco.yaml file in the data folder and name it people.yaml. and redirect the path address in the file (train: ***** val: *****), then change the number of categories (nc) to 1 and finally change the category name to person. Next, copy the YOLO5s.yaml file from the mods folder and paste and rename it to people.yaml. All that needs to be done here is to change the number of nc's in the first line to 1. It is important to note that if the pre-training weights used are YOLOv5s.pt then the corresponding yaml weights call file should also be selected in the mods. YOLOv5 generally has only four weight files (l, m, s, x) and this project uses s.

Finally, the direction path in the train.py file needs to be changed to

```
parser.add_argument('--weights', type=str, default='yolov5s.pt', help='initial weights path')
parser.add_argument('--cfg', type=str, default='models/yolov5s-person.yaml', help='model.
Yaml path')
```

```
parser.add_argument('--data', type=str, default='data/people.yaml', help='data. yaml path')
It also allows a choice of whether running with a GPU or a CPU and the number of cores the CPU
is involved in the calculation. Next it is necessary to download and install a copy of Microsoft
visual C++ and thus it is possible to start training. During the training process the model can be
viewed in real time using TensorBoard, just open a web browser and type in localhost:6006.
```

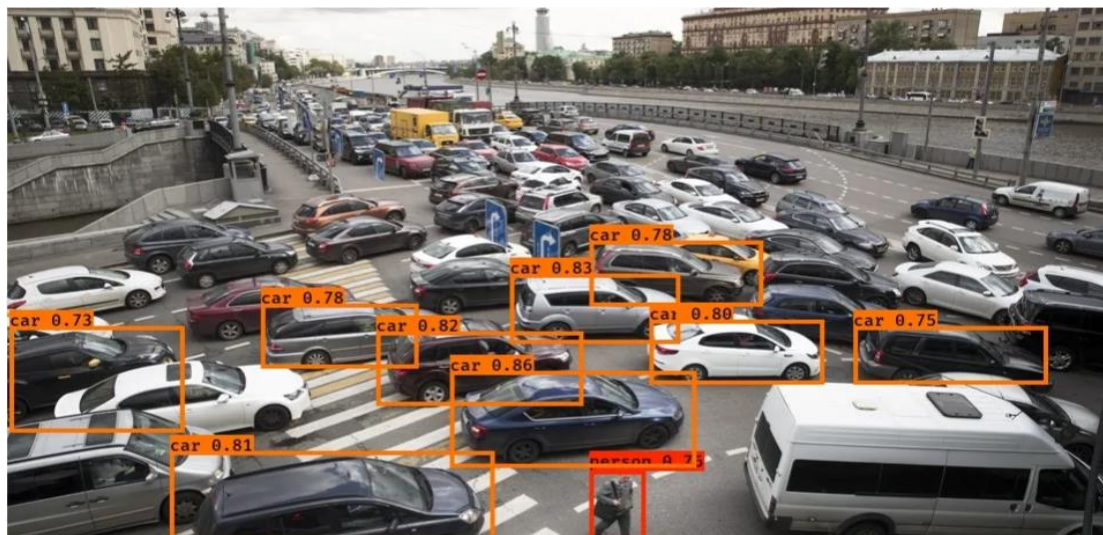
7.1.5. Detect test

After training the model, two weight files and a record of the training results will be obtained, one for last.pt and one for best.pt. Last.pt is the latest round of training and best.pt is the best of all results. Once this weight file is obtained, parser.add_argument ('--weights', nargs='+', type=str, default='yolov5s.pt', help='model.pt path(s)') in detect.py needs to be changed from yolov5s. pt to the file path of best.pt. And modify the video input path to be detected parser.add_argument ('--source', type=str, default='test3.avi', help='source'). Here the input source can be set to 0, which means that the camera input is used. Some of the configuration is the same as for the training, with the option to use the GPU or CPU, and the option to save the detection data or not. For this project, the camera was first used for testing. In the experiments, I acted as the subject and experimented with different angles.

7.1.6. Confidence level analysis

In the image it can be seen that the different objects are labelled and the value to the left of the

label is the confidence level. The confidence score indicates how certain the model is about the object contained in the box and how accurate it thinks the box is in predicting it. The confidence score can be calculated using the following formula: $C = \text{Pr}(\text{object}) * \text{IoU}$. IoU: the concatenated intersection between the predicted box and the ground truth. If no object exists in the cell, its confidence score should be zero. (Shivaprasad, P.2020)



In the analysis of the confidence level, the model trained using the dataset I collected did not have a high confidence level for human recognition. At this point there were only 96 images in the dataset and 300 rounds of training had been carried out. Of course, the problem becomes clear in the iterative analysis of the test results. The confidence level stays on average between 0.4 and 0.6, rarely reaching 0.8, and only when the face is at a suitable angle and facing the camera. It is only when the face is at a suitable angle and facing the camera that the confidence level rises to between 0.7 and 0.8. The problem that must be considered is that this project was developed to help with fire evacuation, so the camera angle deployed for this program lock will be higher up in the room. However, in tests on the higher angles, the confidence level averages were also unsatisfactory. [\[appendix first time\]](#)

7.2. Second stage

7.2.1. Try to improve confidence level

After analysing the first confidence analysis, an attempt to increase the confidence level was started due to the low confidence level and the lack of accurate detection. In the second attempt, the initial dataset was extended from 96 photos to around 140 photos. And the extended part included some different angles and lighting environments. The reason for only training 100 rounds was to avoid long training sessions and still getting a bad result. The second attempt was neither a failure nor a success.

On the third attempt, the number of training rounds was increased according to the results of the second attempt, and the number of training rounds was set to 100,000. However, this time the training also failed, first at 40002/99999. This was due to some unexpected reasons, such as high hardware temperatures. Even though the training was terminated near the general time,

the two weight files were generated. However, this did not change much and the results were still unsatisfactory.

For the fourth attempt, the MOT dataset was used. and transformed the dataset format and performed 300 rounds of training to get a more satisfactory result.

7.2.2. Second stage Confidence level analysis

For the analysis of the second attempt:

After observing the playback video, it was concluded that the confidence level did not change much from the first training. There was only a slight increase in confidence, and such a small increase is negligible. So, the average confidence level was also between 0.5 and 0.6. But at this point a new problem arose, the detector started to recognise some clothes or bed sheets as human. This even seemed a bit scary and bizarre when doing this analysis. Of course, the reason for this is mostly due to the number of times the dataset was trained and the quality of the dataset. [\[appendix second time\]](#)

For the analysis of the third attempt:

After observing the playback video of the third attempt, it was concluded that the confidence level did not improve, but rather decreased slightly. However, there was an improvement in the case of false detections of objects that would have a human-like shape. In this experiment, the confidence level for myself dropped to a minimum of around 0.25 and reached a maximum of around 0.8, with an average value between 0.4 and 0.5. So, such results were clearly not as expected, and after rigorous analysis, one key factor affecting performance was identified. Because each label in the dataset was manually labelled by myself, the coordinates of these labels were not very strict and could even be described as somewhat casual. [\[appendix third time\]](#)

Regarding the analysis of the fourth attempt:

After observing and analysing this attempt it was concluded that the confidence level was significantly improved. And the average confidence values also remained within a stable range under testing of subsequent multi-person environments. So, a more rigorous and standardised dataset is indeed the key to improving detection accuracy. [\[appendix fourth time\]](#)

7.3. Third stage

7.3.1. Collaborate with DeepSort tracking

After completing the above section, it is time to start downloading the DeepSort source code, which is available with a license from GitHub. deep sort is one of the more commonly used

algorithms for multi-target tracking. DeepSort is based on sort target tracking and introduces an apparent feature extraction network, which is trained on a pedestrian re-identification dataset. This network is trained on a pedestrian re-recognition dataset and extracts the apparent features of the target for nearest neighbour matching during the target tracking process, which can effectively improve target tracking in the presence of occlusion. The algorithm can be divided into four core components: trajectory processing and state estimation, relevance metrics, cascade matching, and a depth feature descriptor. The trajectory processing and state estimation part of the algorithm is similar to the sort algorithm and is divided into motion state estimation and target creation and removal. (Sharpless.2021)

DeepSort works in the following way.

1. A YOLOv5 pre-trained model is used to detect the target in the video or image and obtain the target's bounding box and category information.

DeepSort uses Kalman filtering to track and re-identify the target using appearance features to improve tracking accuracy.

3. Along with DeepSort tracking, strategies are used to consider the motion and spatial relationships of the target in the video sequence to further improve tracking results

A source code that already implements the detector interface call has been chosen here. This saves some time in writing the code for the YOLO and deep sort calls. After the download is complete the trained YOLO weight files need to be placed in the weights folder of deep sort. Next create a run.py file in the root directory of the code.

7.3.2. Detectors and trackers call and write counters

Using import to invoke and reference the required files and code within the newly created py file, first create the class for the detector, then create the class for the detector backbone network, and then integrate the output of the backbone network. Next four variables need to be created, the

```
list_overlapping_blue_polygon = [ ]
list_overlapping_yellow_polygon = [ ]
down_count = 0
up_count = 0
```

Then use the two codes `font_draw_number = cv2.FONT_HERSHEY_SIMPLEX`, `draw_text_postion = (int(960 * 0.01), int(540 * 0.05))` these variables are displayed and then have them displayed in the appropriate positions. Then write the rules for calculating where the bbox triggers the bump. for `item_bbox` in `list_bboxes`: `x1, y1, x2, y2, label, track_id = item_bbox`. The human world is a 3-dimensional world, where there is a relationship between object proximity and perspective. When an object is further away from the camera then the object will be smaller, when an object is closer to the camera then the object will be larger. When implementing tracking some people suddenly appear on the screen from certain corners, so here you need to set a code to offset the crash point.

```
y1_offset = int(y1 + ((y2 - y1) * 0.9))
x1_offset = int(x1 + ((x2 - x1) * 0.1))
y = y1_offset
```

`x = x1_offset`

Once this function has been implemented it is necessary to make a determination of the crash line, which is just a metaphorical description. The essence of this function is simply to determine the zone in which the Bbox overlaps based on the track ID. If the overlap occurs first in the blue zone and then in the yellow zone, then the variable `down_count` is added by one and vice versa for `up_count`. The statistics should be displayed in real time while the count is being implemented. Of course this project is ultimately about achieving the current number of people in the room, so adding a `str(down_count-up_count)` when writing the `text_draw` code will do the trick.

7.4. Final

7.4.1. Design interactive interface

As this project needs to be considered for ease of use, it is important to design an interactive interface that is easy to use. The code for this part is continued in the `run.py` file. Create a class for the ui, add the functions that need to be used such as selecting files, exiting etc. The interface will also include a video display and a live message display. Once these text boxes and buttons are set up, it need to bind the events that will be triggered by clicking on them.

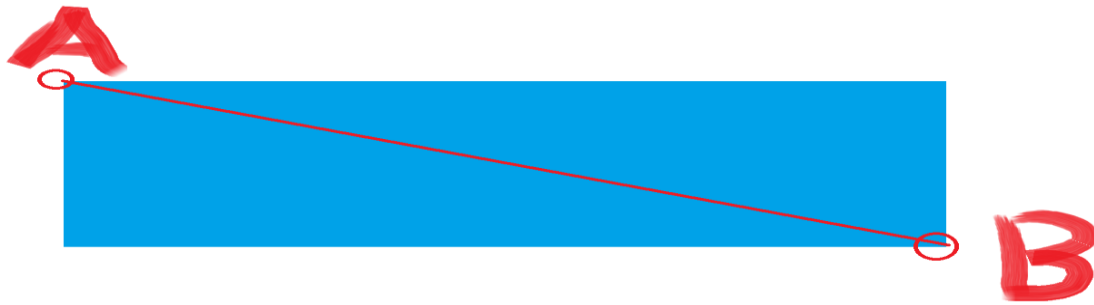
The next step is to add a simple login screen, which contains two input boxes for the account and password respectively, and then a login and registration button, creating a folder named `user` in the root directory to which the registered accounts will be directed. This is because if there were to be some validation mechanism such as a confirmation code, then a server would need to be deployed, whether it be an SMS confirmation code or an email confirmation code, an additional service would need to be deployed to send the confirmation code, which is not necessary for this project as it stands. The login button simply determines whether the input matches the contents of the user folder and logs in, if not then an error value is returned in the input box.

7.4.2. Optimizing calls and counting

During the development process, some problems were discovered. Due to the different size of the input source video, sometimes the video display area of the UI interface does not fully display the video, so in this case the size of the input file should be scaled. In simple terms, this means modifying the resolution of the input source. Here, the resolution has been standardised to `960*540`.

Due to the different positions and angles of detection, the yellow and blue judgement zones cannot be fixed. To modify the position of the judgement zone, the code needs to be modified manually, which of course undermines the principle of ease of use that the project adheres to. So, a point selection function has been added to change the position of the judgement zone by simply marking four points in the pop-up image. The first and second are blue areas and the third and fourth points are yellow areas. As the judgement area can only be a rectangle or square,

the first and second points will act as the maximum diagonal of the rectangle, Same for the third and fourth points. This is shown in the following diagram:



However, the yellow and blue judgement areas were not scaled because the point selection function was added, so the point selection function has been optimised to scale the selected area as well. However, this does not solve the problem completely and the yellow and blue areas are still offset to some extent. This is most likely due to the fact that it is called after scaling. In this case, we can only add a few lines of code to force it to adapt to the 960*540 resolution.

```
list_bboxes = [ ]
    print(list11)
    xx1 = int(list11[0][0] / 1.8)
    yy1 = int(list11[0][1] * 1.8)
    xx2 = int(list11[1][0] / 1.8)
    yy2 = int(list11[1][1] * 1.8)

    xx3 = int(list11[2][0] / 1.8)
    yy3 = int(list11[2][1] * 1.8)
    xx4 = int(list11[3][0] / 1.8)
    yy4 = int(list11[3][1] * 1.8)
```

A clear comparison of the results can be seen in the appendix [\[Appendix Comparison of Effects\]](#)

To improve the usefulness of the result area below the video area two lines of code are added to the code that determines the crash line, under the lines `up_count += 1` and `down_count += 1` respectively.

```
ui.printf(f'label: {label} | id: {track_id} | up way | amount of up: {up_count} | up way id list: {list_overlapping_yellow_polygon}')
ui.printf(f'label: {label} | id: {track_id} | down way | amount of down: {down_count} | down way id list: {list_overlapping_yellow_polygon}')
```

Due to the addition of these two real-time refreshed texts, it is necessary to optimize the ID list and remove useless IDs. If not deleted, the IDs in all records will be displayed every time. This will take up a lot of display space and is not conducive to reading. [\[Appendix Clean ID list\]](#)

8. Test and result analysis

8.1. White box test

Login interface:

Run the program run.py to enter the login screen. The font in the welcome window is displayed normally, the font in the two input boxes is displayed normally, and the registration and login button fonts are displayed normally. The imported white background is applied.

Using the correct values, the user and password are directed to the main windows and the value welcome user is printed in the message area. this means that the code for the login screen directed to the function screen is working.

Test entering the account and password separately using the incorrect values, after entering the wrong account the account text box position returns the value { 'account error'}. After entering an incorrect password, the password text box position returns the value { 'passwd error'}. When a different error value is entered, only the corresponding text box returns the error value. This is as expected. However, if both the wrong account and password are entered, only the value { 'account error'} is returned, which is where this code falls short.

Logging in with the administrator account (account: XQ123 password: 123) successfully takes you to the function screen. All user information values are printed in order in the information area. This means that the code set by the administrator is fine and that the path to the user information is valid and that the user information is actually being determined on the login screen, rather than directly using the account and password pre-set by the code

Enter a combination (000,000) in the account and password fields that has never been registered before. In the background you can see that the user information has been imported into the user folder. Then try the password or account value is empty or both, all three cases can still be registered and the information imported back into the user folder. The administrator account judgment rule is that if the first two letters of the account are XQ, it is determined that this account is an administrator. Then, during registration, it is attempted to discover that the administrator can be registered as long as the first letter is XQ. But if XQ is not in the first and second positions of the text, it cannot be considered an administrator. Another thing to note is that if the same account name is registered repeatedly, the same account already in the user folder will be overwritten. This is a minor bug and more rules should be added to limit the input of the registration function.

Function screen:

After logging into the main function screen, the title value, reminder value and message area value are all displayed in the appropriate places, as well as the correct window text.

After clicking on the Select File button the Windows Explorer window pops up and after selecting the appropriate file the file path is correctly printed out in the message area. So the value of the code ('The currently selected file path is: %s" % filepath') is valid, and it is necessary to wait for the Run button to be clicked, so no other programs should be run, as expected. At this point the file is selected once and the new file path is also displayed.

Clicking the Run button without a file selected ends the program directly, which is expected as the code is set up to end thread 2 if thread 1 is not created. Running with a file is normal, firstly the interface for the point selection function pops up, after selecting the interface the point selection window is closed and monitoring begins, if the process ends directly without selecting a point or if the point position does not meet the requirements for determination. The program also runs normally if in the case of selecting a file is replacing the first selected file. This means that thread 1 was effectively created and called by thread 2. However, if the Select Video button is clicked here while the program is running and a video is selected, the new video is monitored first and the first video is monitored after the new video is finished. This is not a bug, it is a feature that can be exploited in future developments where multiple videos will be monitored simultaneously.

The exit button is not a problem in any case, and the process can be ended at any time. However, there is a problem in that clicking the close button directly while the program is running (while the video is being monitored), the program will continue to monitor until the video ends, which is a bug, and an event binding to the window close button needs to be added to the loop in the main function screen to give the button the ability to end the thread directly.

8.2. Black box test

Firstly, the login and registration functions are available in the first screen and the basic requirements are implemented in the main screen, where the three buttons are for selecting a file, running, and exiting. After following the relevant prompts, the program implements a machine vision-based headcount of the room. The real-time headcount can be seen in the top left corner of the video display and the real-time information can be seen at the bottom of the video display. However, the program does not implement multiple video detection and the monitoring source is not a CCTV camera. Of course these can be developed in the future, as mentioned earlier, due to various resource constraints the project can only reach this level at the moment.

8.3. Performance analysis

The performance of the project is evaluated mainly on the accuracy of the counts, which are concluded after several tests using the same video. The selection of points in the decision area and the angle of the video will directly affect the accuracy of the count. Firstly, the selection of points in the judgement area, which is included in the judgement conditions mentioned above, sets the offset of the judgement points. The reason for the offset is that when an object is detected, a bounding box is drawn outside the outline of the object, and the detection point is set in the top left corner of the bounding box, as this is normally the position where the top left corner does not overlap with other objects. However, when a person is moving the upper left corner of the bounding box is likely to be skipped over by the determine area and not overlapped. There is also the issue of angle. If the camera is deployed in person, the statistics will be absolutely accurate, excluding unusual circumstances. So, the project is meeting the performance requirements, but only if these two key premises are met.

An additional note here is that multi-video simultaneous detection is not an implemented feature. However, this will be explored here. Firstly, there is a question about the performance of the graphics card if multiple videos are to be detected at once. The graphics card on the test PC (RTX 3070 Ti Laptop) was already at 50-60% when monitoring a single video, which frankly isn't the best, but it's still a good performer at this time. If a single video is monitored at such a rate, what if there are dozens or even hundreds of images. So, we need to optimise the code for future development of multi-video monitoring, and we are looking forward to more technological breakthroughs in the chip manufacturing industry to produce higher performance graphics cards.

9. Project evaluation and conclusion

Looking back at the entire design and development process, one has to admit that the project has been challenging. However, it is pleasing to note that the project has achieved 95% of the goals initially set. There are, of course, some shortcomings and there are many reasons for these shortcomings. The key reason is in terms of human and equipment resources. The human resources consist of only one average university student, and it is difficult to afford the expensive equipment and there is no authority to access CCTV videos. If the project is to continue to be developed in the future, we need to focus on solving the problem of program compatibility with CCTV. With more resources, the program would grow rapidly and would only take about two years to develop into a mature product. However, getting to this point will not be easy and we will need to address the issues found in both the black box and the white box. No software in this world is perfect, we can only make it better. In the process of researching and developing this project, I have learned many new things and gained a deeper understanding of artificial intelligence. This study has triggered me to think more deeply about life, society, and the universe. I even look forward to seeing an AI with self-awareness in the future.

Now, AI technology is entering an explosive phase of development. For example, Chat-gpt, a chatbot from OpenAI, has caused a stir in many fields, even to the extent that some countries have explicitly banned the use of the model in their education departments. openAI has also introduced the DALL-E image generation model, GShard, a distributed training technology from Google, Codex, an automated programming tool from OpenAI, Google Brain's Perceiver model and the Switch Transformers model developed by CMU and Facebook AI Research, among others. These are just a few of the AI models that have emerged recently. As technology continues to evolve, more innovative models will emerge in the future. An exciting application of AI in medicine was recently developed by researchers using AI to develop a new antibiotic called halicin. This approach could lead to more medical discoveries in the future and could also be used to discover drugs to treat other diseases, such as cancer and neurodegenerative disorders.

Some are now calling this era the age of artificial intelligence, but we cannot be sure if this is the beginning of the curtain on the age of AI. What is certain, however, is that AI technology will continue to evolve rapidly. As a researcher and developer on this project, I hope that this project will catch up with the times and be able to help more people in the future as it is developed and matured, serving government departments and various organisations to help keep people out of fire.

10. Reference

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11. Appendix

11.1.1. Device info

Host Name: XQ
OS Name: Microsoft Windows 11 Professional
OS Version: 10.0.22000 N/A Build 22000
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: ****
Registered Organization: ****
Product ID: ****
Original Install Date: 2023/3/23, 6:11:49
System Boot Time: 2023/4/15, 18:49:14
System Manufacturer: ASUSTeK COMPUTER INC.
System Model: ASUS TUF Gaming A17 FA707RW_FA707RW
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: AMD64 Family 25 Model 68 Stepping 1 AuthenticAMD
~3201 Mhz
BIOS Version: American Megatrends International, LLC. FA707RW.314,
2022/10/12
Windows Directory: C:\Windows
System Directory: C:\Windows\system32
Boot Device: \Device\HarddiskVolume2
System Locale: ****
Input Locale: ****
Time Zone: ****
Total Physical Memory: 64,765 MB
Available Physical Memory: 55,385 MB
Virtual Memory: Max Size: 68,861 MB
Virtual Memory: Available: 56,524 MB
Virtual Memory: In Use: 12,337 MB
Page File Location(s): ****
Domain: ****
Logon Server: ****
Hotfix(s): 5 Hotfix(s) Installed.
[01]: ****
[02]: ****
[03]: ****
[04]: ****

[05]: ****

Network Card(s): 3 NIC(s) Installed.

[01]: Realtek PCIe GbE Family Controller

Connection Name: Ethernet

DHCP Enabled: Yes

DHCP Server: *****

IP address(es)

[01]: 10.246.200.234

[02]: fe80::8c5:234c:43fa:a671

[02]: MediaTek Wi-Fi 6 MT7921 Wireless LAN Card

Connection Name: WLAN

Status: Media disconnected

[03]: Bluetooth Device (Personal Area Network)

Connection Name: Bluetooth connected

Status: Media disconnected

Hyper-V Requirements: VM Monitor Mode Extensions: Yes

Virtualization Enabled In Firmware: Yes

Second Level Address Translation: Yes

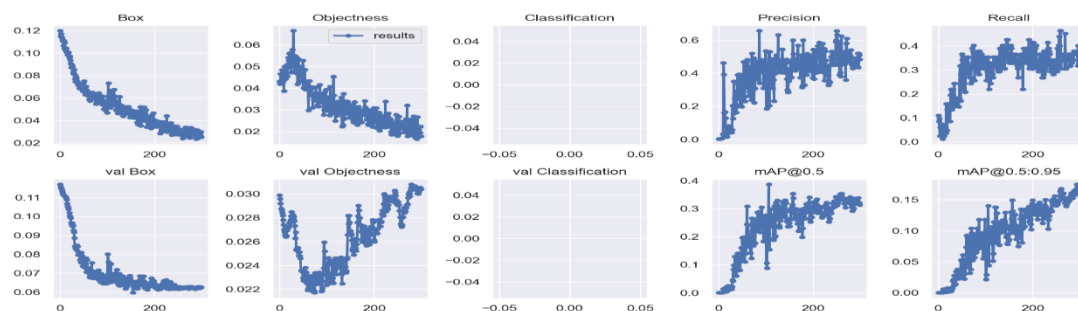
Data Execution Prevention Available: Yes

NVIDIA-SMI 531.41			Driver Version: 531.41		CUDA Version: 12.1		
GPU	Name	TCC/WDDM	Bus-Id	Disp.A	Volatile Uncorr. ECC		
	Fan Temp	Perf	Memory-Usage		GPU-Util	Compute M.	MIG M.
	Pwr:Usage/Cap						
0	NVIDIA GeForce RTX 3070 T...	WDDM	00000000:01:00.0	On	N/A		
N/A	51C	P0	1108MiB / 8192MiB		1%	Default	N/A

11.1.2. Confidence level analysis

first time (96 images, 300 times) ([click for going back to article](#))

Training results

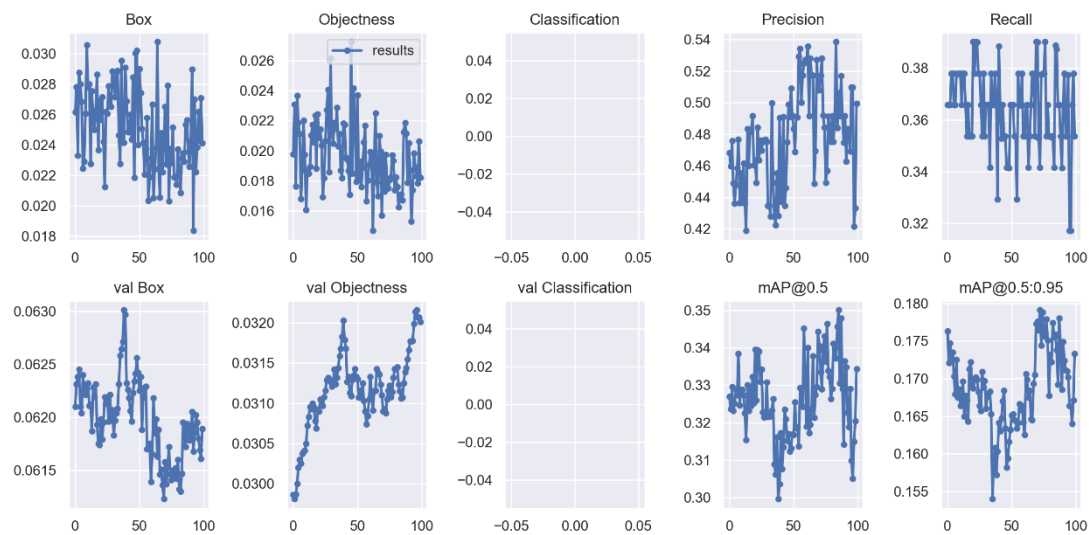


detection result



second time (136 images, 100 times) [\(click for going back to article\)](#)

Training results



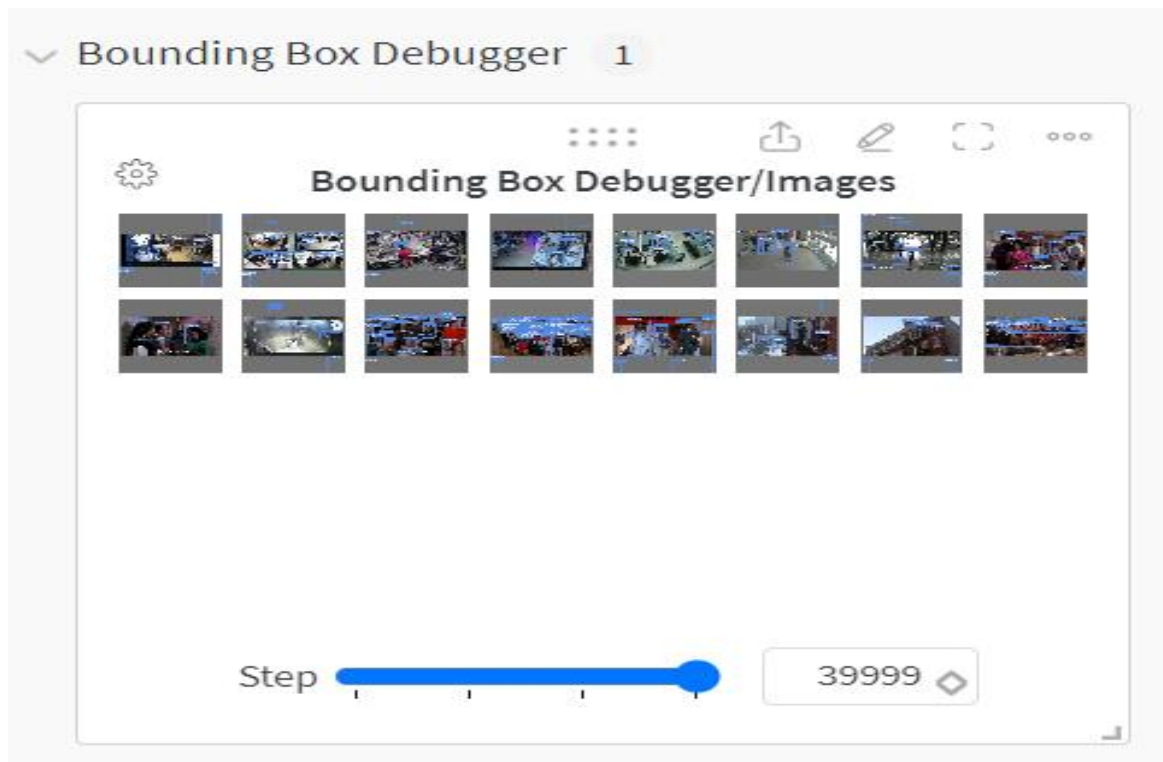
detection result



third time (136 images, 40000 times) [\(click for going back to article\)](#)

Training results





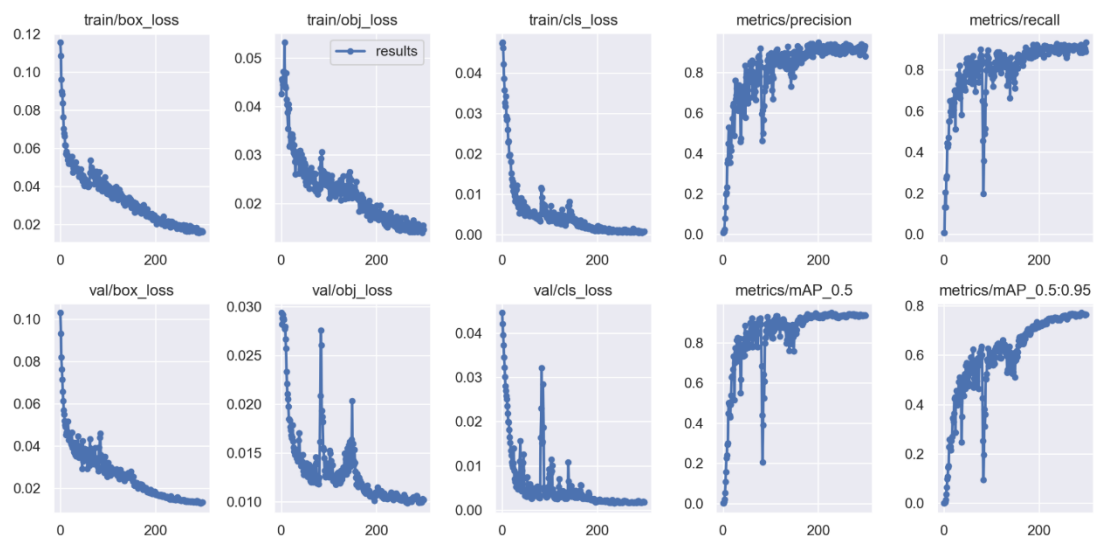
detection result





Fourth time (MOT, 300 times) [\(click for going back to article\)](#)

Training results



11.1.3. Comparison of Effects

[\(click for going back to article\)](#)



11.1.4. Clean ID list

[\(click for going back to article\)](#)

```
# ----- Clear useless IDs-----  
list_overlapping_all = list_overlapping_yellow_polygon + list_overlapping_blue_polygon  
for id1 in list_overlapping_all:  
    is_found = False  
    for _, _, _, _, bbox_id in list_bboxes:  
        if bbox_id == id1:  
            is_found = True  
            break
```

```

        pass
    pass

    if not is_found:
        # if not found, delete id
        if id1 in list_overlapping_yellow_polygon:
            list_overlapping_yellow_polygon.remove(id1)
        pass
        if id1 in list_overlapping_blue_polygon:
            list_overlapping_blue_polygon.remove(id1)
        pass
    pass

    list_overlapping_all.clear()
    pass

# Clear List
list_bboxes.clear()

```

11.1.5. Proposal

COMP [1682] Project Proposal

Final year project

Yuqing Luo

Image Recognition to Aid Fire Evacuations

001262439-X

1 Overview

At present, most object recognition is in the world or for some static objects, and there is no model for indoor fire. This project is for these kinds of situations. So, when a building is evacuated it is possible to know how many people were present in each room. The system would know whether everyone evacuated and could provide an all clear for each monitored room. This model will also have a lot of expansion space in the future. You can judge the fire point and fire spread direction based on its own functions, and even add some other algorithms to judge some other things. For example, whether the site is safe or not, and whether there is any dangerous action. This project will use some open-source software to help complete the project. And there are many ways to achieve my goal in this project. This project will also apply many basic computer technologies, such as digital image underlying logic and computer environment. And some complex computer technologies such as computer vision and artificial intelligence.

Keywords: [indoor fire, monitored room, open-source software, digital image, computer environment, computer vision, artificial intelligence]

2 Aim

This project will be investigating to analyse the number of people in the room, to help people

better escape from the fire and to better assist firefighters. It will be able to count the number of people in the building. And it will use computer vision and artificial intelligence to achieve the goal.

3 Objectives

3.1 Objective Detect people

- 3.1.1 track their movements, and draw a frame
- 3.1.2 building environment [2weeks]
- 3.1.3 coding [2weeks]
- 3.1.4 test [1weeks]

3.2 Objective Identify people

- 3.2.1 Distinguish People and objects and strengthen people characteristics
- 3.2.2 training [3weeks]
- 3.2.3 test [1week]

3.3 Objective Single space count

- 3.3.1 Only calculate the people in a room or hall.
- 3.3.2 coding [3weeks]
- 3.3.3 test [1week]

3.4 Objective Practical Experiment

- 3.4.1 Conduct real simulation test and monitor test in a room.
- 3.4.2 Multiple experiments [1week]
- 3.4.3 documenting a log [0.5weeks]
- 3.4.4 Summary and correction [0.5weeks]

3.5 Objective Total space count

- 3.5.1 Ccalculate all of people whom in the building
- 3.5.2 coding[3weeks]
- 3.5.2 test[1weeks]

3.6 Objective Practical Experiment

- 3.6.1 Conduct real simulation test and monitor test in a room.
- 3.6.2 Multiple experiments [1week]
- 3.6.3 documenting a log [0.5weeks]
- 3.6.4 Summary and correction [0.5weeks]

Objective	Method	Deliverable	Duration
<i>Objective 1</i>	waterfall	<i>Deliverable 1</i>	<i>5 weeks</i>
<i>Objective 2</i>	waterfall	<i>Deliverable 2</i>	<i>3 weeks</i>
<i>Objective 3</i>	waterfall	<i>Deliverable 3</i>	<i>4 weeks</i>
<i>Objective 4</i>	waterfall	<i>Deliverable 4</i>	<i>2 weeks</i>
<i>Objective 5</i>	waterfall	<i>Deliverable 5</i>	<i>4 weeks</i>
<i>Objective 6</i>	waterfall	<i>Deliverable 6</i>	<i>2 weeks</i>

3.7 Project Framework or Any Methodology used

my project following the waterfall method, because every single object is a foundation for next

object in my project. So I have to finish every single by step.

4 Legal, Social, Ethical and Professional

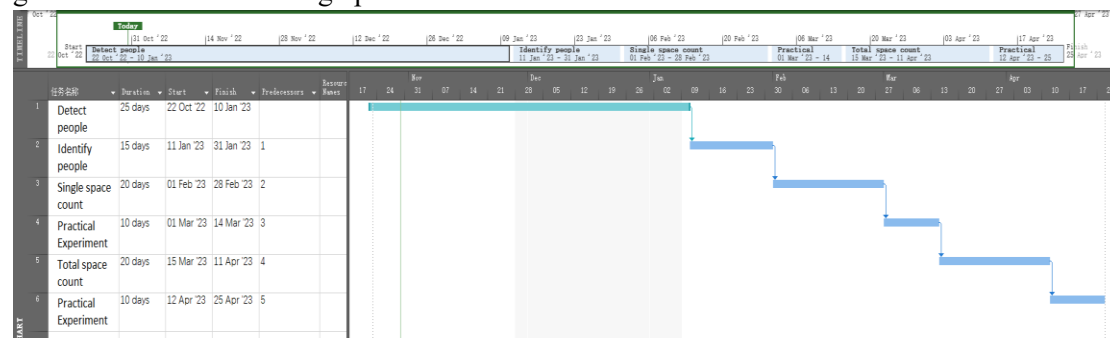
First, regarding the law, the background of this project should be that the number of people in the building cannot be known to prevent fire, so the camera will certainly be installed in the public area, so it will not infringe the portrait right. However, data (such as videos or photos) may be exposed, which will violate the “Data protection Act 2018”. At the same time, if the deployment of the camera in terms of network security is defective or illegal operation by internal personnel through convenience) , it may also violate the “Computer Misuse Act 1990”.

The core purpose of this project is to help people stay away from fire better, so this project is beneficial to society.

In the process of project development, we should pay attention to whether some things are open source, or when we want to borrow other people's code, we need to win their consent. In addition, in the development process, it is required to comply with the standards of some professional groups, such as IEEE, ACM, CCF.

5 Planning

At first, you need to integrate your own knowledge, and then think about where to start. This project will involve a lot of computer vision and neural networks, as well as high-level language programming. And many different libraries will be used. Later, I will learn and practice to confirm the best way to complete this project. This project should be completed in more than one way. There will be many practical models, such as VGG, ResNet Faster R-CNN, YOLO Deep Lab, U-NET Show and Tell GAN So for me, my method is to learn while practicing. And get the best solution through practice.



6 Initial References

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Zhu, Z. and Cheng, Y. (2020) Application of attitude tracking algorithm for face recognition based on OpenCV in the intelligent door lock. UNKNOWN, UNKNOWN: Computer Communications.

Sainath, R.M. et al. (2022) Comparative Performance Study of CNN-based Algorithms and YOLO. UNKNOWN, UNKNOWN: Computing and Communication Technologies (CONECCT).

Sharma, S. (2022) in Human Face Mask Identification using Deep Learning with OpenCV Techniques. 2022 7th International Conference on Communication and Electronics Systems (ICCES). Available at:

[https://librarysearch.gre.ac.uk/client/en_GB/gre/search/detailnonmodal/eds:\\$002f\\$002f-1412568528\\$002f0\\$002fedsee\\$007c\\$007cedsee.9835884/ada/330c017c-611d-4144-b07b-34e975b8c164.vi61LBUBq4TyRV3MewmG3Q\\$003d\\$003d?qu=Human+Face+Mask+Identification+using+Deep+Learning+with+OpenCV+Techniques%23&if=el%09edsSelectFacet%09FT%7C%7CRV&if=ee%09edsExpanderFacet%09fulltext&d=eds%3A%2F%2F-1412568528%2F0%2Fedsee%7C%7Cedsee.9835884%7E-1412568528%7E0&ir=Both&h=8](https://librarysearch.gre.ac.uk/client/en_GB/gre/search/detailnonmodal/eds:$002f$002f-1412568528$002f0$002fedsee$007c$007cedsee.9835884/ada/330c017c-611d-4144-b07b-34e975b8c164.vi61LBUBq4TyRV3MewmG3Q$003d$003d?qu=Human+Face+Mask+Identification+using+Deep+Learning+with+OpenCV+Techniques%23&if=el%09edsSelectFacet%09FT%7C%7CRV&if=ee%09edsExpanderFacet%09fulltext&d=eds%3A%2F%2F-1412568528%2F0%2Fedsee%7C%7Cedsee.9835884%7E-1412568528%7E0&ir=Both&h=8) (Accessed: October 22, 2022).

11.1.6. Open-source software and dependent libraries

[\(click for going back to article\)](#)

Software

Anaconda==2023.3

PyCharm==2022.3.3

LabelMe

YOLOv5==3.0

DeepSort

Cuda+cuDNN

Dataset

MOT challenge==MOT20

Additional need of operating environment

Python==3.7

Microsoft visual C++

Dependent files

Download address:

1. Official website of various software
2. Anaconda default address
3. Pip default address
4. Tsinghua Mirrors ---- <https://mirrors.tuna.tsinghua.edu.cn/>
5. GitHub

Package	Version
-----	-----
absl-py	1.4.0
altgraph	0.17.3
cachetools	5.3.0
certifi	2022.12.7
chardet	4.0.0
charset-normalizer	3.1.0
colorama	0.4.6
cycler	0.11.0
dataclasses	0.6
easydict	1.10
fonttools	4.38.0
future	0.18.3
google-auth	2.17.2
google-auth-oauthlib	0.4.6
grpcio	1.53.0
idna	3.4
imageio	2.27.0
importlib-metadata	6.1.0
imutil	0.3.4
kiwisolver	1.4.4
Markdown	3.4.3
MarkupSafe	2.1.2
matplotlib	3.5.3
networkx	2.6.3
numpy	1.21.6
oauthlib	3.2.2
opencv-python	4.7.0.72
packaging	23.0
pandas	1.3.5
pefile	2023.2.7
Pillow	9.5.0

pip	22.3.1
protobuf	3.20.3
pyasn1	0.4.8
pyasn1-modules	0.2.8
pyinstaller	5.10.1
pyinstaller-hooks-contrib	2023.2
pyparsing	3.0.9
PyQt5	5.15.9
PyQt5-Qt5	5.15.2
PyQt5-sip	12.11.1
python-dateutil	2.8.2
pytz	2023.3
PyWavelets	1.3.0
pywin32-ctypes	0.2.0
PyYAML	6.0
requests	2.28.2
requests-oauthlib	1.3.1
rsa	4.9
scikit-image	0.19.3
scipy	1.7.3
seaborn	0.12.2
setuptools	65.6.3
six	1.16.0
tensorboard	2.11.2
tensorboard-data-server	0.6.1
tensorboard-plugin-wit	1.8.1
thop	0.1.1.post2209072238
tifffile	2021.11.2
torch	1.7.0+cu110
torchvision	0.8.1+cu110
tqdm	4.65.0
typing_extensions	4.5.0
urllib3	1.26.15
Werkzeug	2.2.3
wheel	0.38.4
wincertstore	0.2
zipp	3.15.0